A Certified 2.68 TeV Gap in the Closed-Form Map of the Compactification Scale in SU(7) Grand Gauge-Higgs Unification, and the Invariant Its Two Observables Cannot See (Part VIII)

High-energy physics · grand gauge-Higgs unification

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Statements

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At a glance

What the paper contributes

- **The inverse map is decidable** on the eight-generator lattice of bulk contents of the SU(7) model of Komori and Maru: a target box in $(1/R_5, m_h)$ pulls back to finitely many cells and, on each, one interval of the moment G (Proposition 1); a cell is finite because only $\mathbf{7}(+,+)$ has $A_4 = 0$ (Observation 1 of the paper).
- **"No content" carries a named certificate**: cone, congruence, an exact rational Farkas pair, exhaustion, or floor. One dual certificate is printed in full.
- **A certified gap of the closed-form map**: on the published seed the enumerated image is 35 points in $[8897.5, 9967]$ GeV ($8D = 1$) and 65 in $[5258, 6215]$ ($8D = 3$), and nothing lies in $(6.216,\ 8.897)$ TeV; at the measured Higgs mass one point, $1/R_5 = 8967.5$ GeV, carried by 81 contents with the same one-loop potential.
- **What the observables cannot see**: on a fixed rung $(1/R_5, m_h)$ fix four of the five complete invariants (Proposition 3), but two contents share both and disagree about the vacuum invariant W.
- Scope stated by the paper: every emptiness is a statement about the closed form, not yet the untruncated model; the recursion of Proposition 2 is classical (Beck–Robins).

🚀 **Start here: Proposition 1**: on a cell with 8D > 0, a target box pulls back to one interval of G, with both ends evaluations.

Main results:

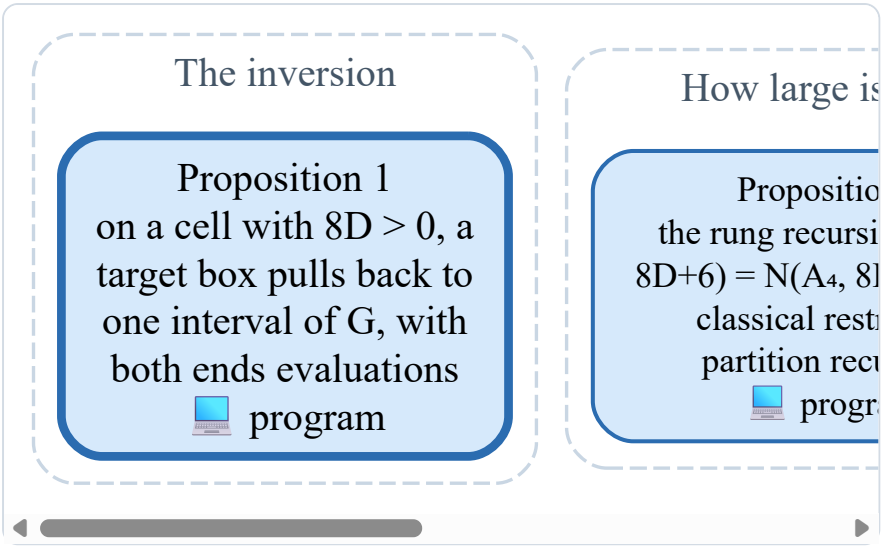
- **Proposition 1**: on a cell with 8D > 0, a target box pulls back to one interval of G, with both ends evaluations
- **Proposition 3**: on a fixed rung the two observables fix A_4, G and (U, V), and leave only W free

3 results | ✅ **0** formalized in Lean | ◐ **0** with a part in Lean

💻 **3** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



1 The inversion

Proposition 1. (the G-window) p. 7

Fix a cell $(A_4, 8D)$ with $8D>0$, and move along the curve (2)(II). Then

$$\frac{dG}{dx} = \frac{A_4}{x} - \frac{12\zeta(3)\,(8D)/8}{x^3} = \frac{A_4x^2 - x^2(6\mu + A_4)}{x^3} = -\frac{6\mu}{x} < 0,$$

so G is strictly decreasing in x. The admissible x are themselves an interval, because solving (2)(II) for μ gives

$$\mu(x) = \frac{\zeta(3)\,(8D)}{4x^2} - \frac{A_4}{6}, \qquad \text{so} \qquad \frac{d\mu}{dx} = -\frac{\zeta(3)\,(8D)}{2x^3} < 0,$$

which makes $\{x>0 : \mu(x) \in [\mu_-, \mu_+]\}$ an interval; intersecting it with $[x_-, x_+]$ leaves an interval $[x_{\min}, x_{\max}]$ when that intersection is non-empty. Hence the set of G a target box admits on this cell is the interval $[\,G(x_{\max}),\ G(x_{\min})\,]$, and both ends are evaluations. When the intersection is empty the cell admits no G at all, and the target has no content there — which is a decision and not a failed search, by §4.

NEW **Classical and new:** the paper lists the exactness of the inversion among what is ours (§1, item 1); the identities (I), (II) are Part VII's.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the inverse map on the published seed: the five certificates, the clusters, the gap, the worked Farkas certificate, the design table; finiteness of a cell (Observation 1) used throughout. `scripts/inverse_design.py`
- **Program:** the same on both gauge seeds in doubled coordinates. `scripts/reachable_set.py`
- **Program:** every interval printed as empty is contained in the one the run certifies. `scripts/check_intervals.py`

- **Program:** the displayed formulas against their Python reading and the archived run; each link perturbed separately and required to fail. `scripts/check_formulas.py`, `scripts/falsify_formulas.py`

2 How large is a cell

Proposition 2. (the rung recursion, and it is the classical one) p. 10

> *Write $P(T, Q)$ for the number of contents built from the seven generators with $A_4^{(j)} \geq 1$ alone that reach $(T, Q) = (A_4 - A_4^{\text{gauge}},\ 8D - 8D^{\text{gauge}})$. Then*
>
> $$N(A_4,\ 8D + 6) \ = \ N(A_4,\ 8D) \ - \ P(A_4 - A_4^{\text{gauge}},\ 8D - 8D^{\text{gauge}}).$$

> NEW **Classical and new:** not claimed: the classical restricted-partition recursion [BeckRobins] in its vector form; the paper claims only the measurement that the correction term is 1 or 2 (§9).

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the counting table by dynamic programming; the recursion checked on 4694 grid points; the four enumerator totals reproduced exactly (69 million contents). `scripts/rung_census.py`

3 The reachable set on the published seed

Proposition 3. (what the two observables fix on a rung) p. 15

> *Fix $8D$. Two contents with the same $(1/R_5, m_h)$ have the same A_4, hence the same G, hence the same (U, V). So at fixed $8D$ the two observables determine four of the five invariants — A_4, $8D$, $2U$, V — and leave only W free.*

> NEW **Classical and new:** the arithmetic is unique factorisation and "is nobody's"; what the paper claims is noticing that the census of the measured-mass fibre depends on it (Acknowledgements, §9).

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** distinct (2U, V) = distinct G on all four rungs (3029, 8769, 16938, 28601); no cross-rung collision among the 314 points; the degenerate pair 2W = +5 / −27 and its exact-potential split of 3.8 MeV. `scripts/what_the_observables_miss.py`

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